



Assignment 3

Course	Artificial Intelligence
Instructor	Ms. Rushda Muneer
Section	BCS-6B, BCS-6C & BCS-6G
Marks	50

Important Guidelines:

- 1. The assignment is hand written. All questions should be solved in order clearing indicating the question number and part.*
- 2. You must submit the soft copy (scanned pdf file) on google classroom.*
- 3. Copying and plagiarism will result in zero marks.*
- 4. There will be no extension for deadline (10th May, 2026 11:59pm)*
- 5. No late submissions are allowed*

Backpropagation

Question 1:

(6+4+6=16)

A neural network has:

- Inputs: $x_1=2$, $x_2=1$

Hidden layer: 2 neurons with ReLU activation

Output layer: 2 neurons with tanh activation

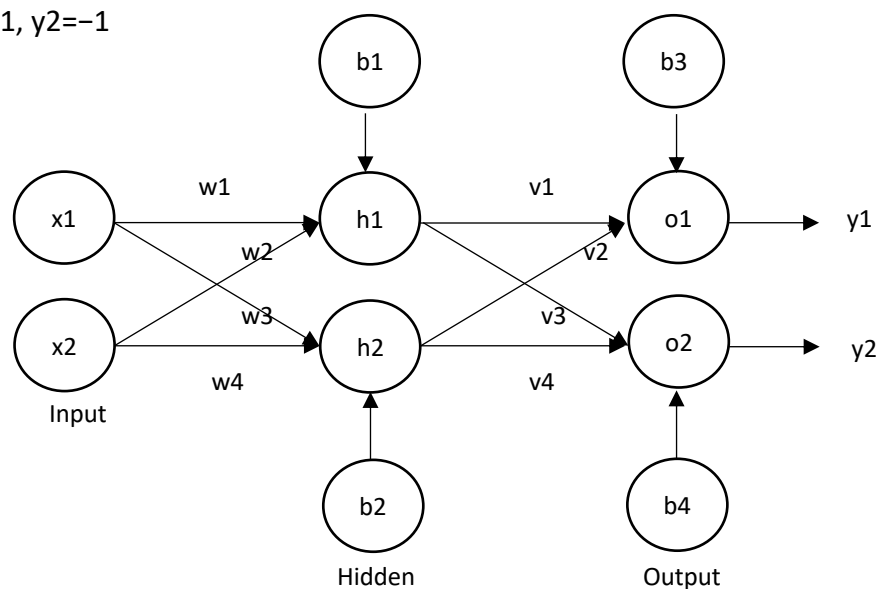
Learning rate: $\eta=0.1$

Initial Weights and Biases

- Neuron h1: $w_1=1$, $w_2=1$, $b_1=1$
- Neuron h2: $w_3=2$, $w_4=-1$, $b_2=0.5$
- Neuron o1: $v_1=1$, $v_2=1$, $b_3=0$
- Neuron o2: $v_3=-1$, $v_4=2$, $b_4=-0.5$

Targets

- $y_1=1$, $y_2=-1$



TASKS

(a) Forward Propagation

Compute:

- Hidden layer outputs
- Output layer pre-activations
- Final outputs

(b) Calculate errors and Gradients

- Output layer neurons
- Hidden layer neurons

(c) Update weights and biases

Clustering

Question 2:

(5+4+3 =12)

A data science team at an e-commerce company is analyzing customer purchase behavior to design targeted marketing strategies. Each customer is represented by their average monthly spending in two categories: Electronics and Clothing.

Customer	Electronics	Clothing
A	2	4
B	3	5
C	8	7
D	9	6
E	10	8

The team applies Agglomerative Hierarchical Clustering using Euclidean distance and Average Linkage.

Tasks

(a) Distance Computation and Analysis

- Compute the full pairwise Euclidean distance matrix.

(b) Hierarchical Clustering

Perform **step-by-step agglomerative clustering** using **average linkage**:

- Clearly show the merge(s)
- Update cluster distances after each merge using average linkage
- Show all intermediate calculations

You must clearly state:

- Which clusters merge at each step
- The computed linkage values at each step
- Any tie-breaking decisions (if applicable)

(c) Dendrogram Interpretation

Draw or describe the structure of the dendrogram formed from the clustering process. Clearly indicate:

- Lifetime at which each merge occurs
- Number of clusters that should be meaningful for the business decisions regarding customer behavior

Convolution

Question 3:

(8+4+2=14)

You are developing a CNN-based system for a self-driving car to classify road signs using two sensor channels:

Channel 1: visual intensity

Channel 2: edge-detected signal

Each input is a 2-channel 4 x 4 image.

Input

- Channel 1:

1	2	0	1
0	1	3	2
2	1	1	0
1	2	2	1
- Channel 2:

0	1	2	1
1	0	1	2
2	1	0	1
1	2	1	0

CNN Architecture

Layer 1: Convolution

- Filters: 2
- Kernel size: 2×2
- Stride = 1
- Padding = 0
- Bias = 0

Filter 1

- Channel 1:

1	0
-1	1
- Channel 2:

0	1
1	0

Filter 2

- Channel 1:

1	-1
0	1
- Channel 2:

1	0
-1	1

Layer 2: ReLU

- $f(x) = \max(0, x)$

Layer 3: Max Pooling

- Pool size: 2×2
- Stride = 1

Layer 4: Flatten

Layer 5: Fully Connected

- Input size = 8
- Output size = 2 (two road sign classes)

Weights: 1 -1 0 1 2 0 -1 1
 0 1 1 -1 0 2 1 -1

Bias: [1,0]

Layer 6: Softmax

(a) Forward Pass

- Compute the output feature maps after Layer 1 (convolution)
- Apply ReLU activation
- Compute output after max pooling
- Flatten the output
- Compute the fully connected layer outputs
- Apply softmax and determine predicted class

(b) Output Dimensions

Compute the output size after:

- Convolution
- Pooling
- Flatten
- Fully connected layer

(c) Trainable Parameters

- Compute parameters in the convolution layer
- Compute parameters in the fully connected layer

Evaluation Metrics

Question 4:

(3+5=8)

A hospital has developed a machine learning model to detect a rare disease. The model is evaluated on 1,000 patients, but due to a reporting issue, the confusion matrix is partially misaligned in documentation.

The reported results are:

- “Correctly identified healthy patients”: 880
- “Incorrectly flagged healthy patients as diseased”: 30
- “Correctly identified diseased patients”: 45
- “Missed diseased patients”: 45

(a)

- Carefully reconstruct the **correct confusion matrix (TP, FP, TN, FN)**.
- State which class is the **positive class** in this medical problem and justify your choice.

(b) Using the **correctly interpreted confusion matrix**, compute:

- Accuracy
- Precision
- Recall
- FPR
- F1-score